Enhancing Multivariate Time-Series Forecasting

From Hypervariate to Hyperspectral Graphs

HSG - HyperSpectral Graph

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Spring 2026

Motivation & Problem Statement

Why does MTS Forecasting matter?

- Traffic flow, energy demand, ECG signals, epidemic spread - all are *multivariate*, *correlated*, and *time-ordered*
- Accurate forecasting = efficiency gains, early warning, better planning

Core modeling challenge:

The Dual Dependency Problem

A good model must simultaneously capture:

- **Intra-series (temporal):** patterns within one variable over time
- **Inter-series (spatial):** correlations *across* variables at each instant

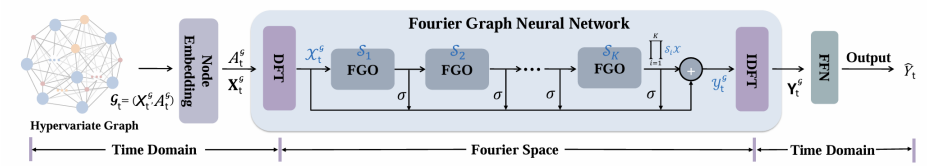
Where classical GNN approaches fail:

- **Two-module designs** (GCN + LSTM) treat space & time as *separate* processes — a fundamental mismatch
- Uncertain compatibility between graph & temporal modules adds *design overhead*
- Separate encoding *violates* the unified spatiotemporal nature of the signal

Our Goal

Can a *pure graph* approach unify spatiotemporal modeling while preserving **temporal order** and **local structure**?

Baseline Recap - The Pure Graph Baseline



FourierGNN pipeline

Key Innovation: The Hypervariate Graph

- Given $X_t \in \mathbb{R}^{N \times T}$, treat each $x_n^{(t)}$ as a node $\Rightarrow NT$ nodes
- Fully-connected adjacency: $A_t^G \in \{1\}^{NT \times NT}$
- Unified spatiotemporal encoding (no temporal module)

Fourier Graph Operator (FGO):

$$\mathcal{F}(X)\, S_{A,W} = \mathcal{F}(AXW)$$

Multi-step prediction:

$$\hat{Y}_t = \mathrm{FFN}\Big(\mathcal{F}^{-1}\big(\sum_{k=0}^{K} \sigma(\mathcal{F}(X_t^G) S_{0:k} + b_k)\big)\Big)$$

Strengths: SOTA (7 datasets), 14.6% faster, 20% fewer params

Limitations:

- Fully-connected graph dilutes local patterns
- Time ordering lost (graph isomorphism)

Two Critical Limitations of the Hypervariate Graph

Limitation 1 - Loss of Sequential Order

A fully-connected graph is permutation-invariant.

Two time series with *identical values* but *different temporal orders* produce **isomorphic** hypervariate graphs - indistinguishable to any GNN.

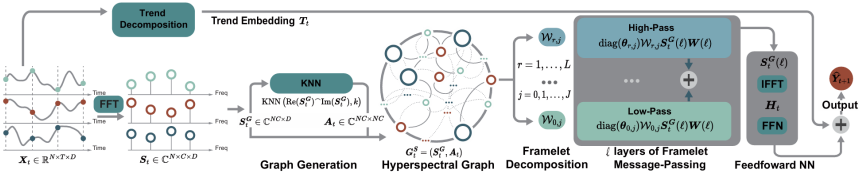
- Permuting up to 100% of time steps in FourierGNN causes only marginal MAE degradation.
- Model *cannot* exploit "what happened before what."

Limitation 2 - Attention Dilution

A uniform $NT \times NT$ adjacency matrix means **all connections are treated equally**.

- Local intra-series patterns (nearby time steps) get same weight as distant inter-series links.
- Tends toward **over-smoothing**: embeddings converge after multiple hops, losing discriminability.

HSG - HyperSpectral Graph Solution



1) Trend Decomposition:

Extract T_t (moving average), operate on residual.

2) Frequency Transform:

$$S_t = \mathcal{F}(X_t) \in \mathbb{C}^{N\times C\times D}$$

Fixes Limitation 1: Fourier coefficients encode ordered temporal structure; permuting the sequence changes the mapping, not the signal.

3) Sparse KNN Graph:

$$A_t = \mathrm{KNN}\big(\Re(S_t^G) \oplus \Im(S_t^G),\ k\big)$$

Fixes Limitation 2: k nearest spectral neighbors $\Rightarrow$ sharper attention, reduced over-smoothing.

HSG - Global Framelet Message-Passing Operator

Framelet Convolution:

Multi-resolution filter banks (low/high-pass)

$$S_t^G(\ell+1) = \mathrm{SiLU}\Big(\sum_{(r,j)\in\mathcal{I}} \mathrm{diag}(\theta_{r,j})\, W_{r,j}\, S_t^G(\ell)\, W^{(\ell)}\Big)$$

- $W_{r,j}$: framelet basis (scale j, filter r)
- $\theta_{r,j}$: learnable spectral weights
- $W^{(\ell)}$: channel mixing matrix

Why Framelets:

- **Global spectral view** - captures long-range dependencies
- **Adaptive filtering:** $\theta > 1$ sharpens, $\theta < 1$ smooths

Output Generation:

$$\hat{Y}_{t+1} = \mathrm{FFN}(H_t) \oplus \mathrm{Lin}(T_t)$$

where $H_t = \mathcal{F}^{-1}(S_t^G(\ell+1))$

Advantage over FGO

Framelets **adaptively balance smoothing and sharpening** across frequency bands, enabling global context modeling without over-smoothing.

Experimental Results: Prediction Length = 12

Dataset	FourierGNN			HSG		
	MAE	RMSE	MAPE (%)	MAE	RMSE	MAPE (%)
ECG	0.052	0.078	10.97	0.061	0.092	15.25
COVID-19	0.123	0.168	71.52	0.141	0.197	**66.42**
SOLAR	0.120	0.162	116.48	0.144	0.191	123.92
TRAFFIC	0.011	0.023	28.71	0.069	0.109	62.03
ELECTRICITY	0.051	0.077	24.28	0.886	7.563	25.09
METR-LA	0.084	0.165	89.69	0.085	0.168	**89.28**
WIKI	0.041	0.076	64.50	0.041	0.294	111.90

Experimental Results: Variable Prediction Length (H)

METR-LA

H	MAE		RMSE		MAPE (%)	
	FourierGNN	HSG	FourierGNN	HSG	FourierGNN	HSG
3	0.050	0.053	0.113	0.116	86.30	**86.29**
6	0.066	0.066	0.140	**0.137**	87.97	**87.42**
9	0.076	0.076	0.159	**0.156**	88.99	**88.47**
12	0.084	0.085	0.165	0.168	89.69	**89.28**

COVID-19

H	MAE		RMSE		MAPE (%)	
	FourierGNN	HSG	FourierGNN	HSG	FourierGNN	HSG
3	0.071	0.079	0.103	0.121	61.02	**56.50**
6	0.093	0.109	0.131	0.156	65.72	**60.41**
9	0.109	0.124	0.148	0.175	69.59	**63.72**
12	0.123	0.141	0.168	0.197	71.52	**66.42**

Conclusion

What We Diagnosed & Fixed:

1. **Sequential order loss** in hypervariate graphs
   $\rightarrow$ Fixed via *frequency domain*: Fourier coefficients inherently encode temporal ordering
2. **Attention dilution & over-smoothing**
   $\rightarrow$ Fixed via *sparse KNN topology* built on spectral features
3. **Short-sightedness in sparse GNNs**
   $\rightarrow$ Fixed via *global framelet message passing* with adaptive low/high-pass filtering

Contributions

- **Omkar Bhandare:** Framelet Messaging Ideation and module implementation
- **Shivam Choudhury:** Hyperspectral Graph ideation and overall implementation
- **Ayush Rawat:** Dataset Sourcing and Pre-processing, Code/Result Organisation